# 교 수 임 용 지 원 서

| 접수번호 | 000011 |
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| 소 속 | 대학 : 공과대학 | 학부(과): 건축공학부 | 전공 : |
| 지원분야 | 지능형 건축환경 시스템 | | |
| 성 명 | (국문) 조현인 | (영문) Hyunin Jo | (한문). 曺賢仁 |

## 1. 인적사항

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 성 별 | 남자 | 국 적 | 대한민국 | 영주권국가 | 대한민국 | 병 역 | |
| 생년월일 | 1994년 08월 22일 (만 31 세) | | | | | e-mail | best2012@naver.com |
| 현 주 소 (국내연락처) | (우편번호 : 072-51 ) | | | | | 전화 번호 | 자 택 : 없음 |
| | 서울 영등포구 영신로40길 20, 805호 (영등포동6가) | | | | | | 핸드폰 : 01063878402 |
| 현근무지 | 현대자동차 R&D 연구소 | 직위 | 책임연구원 | | | | 직 장 : 없음 |

## 2. 학력사항

| 구분 | 입학년월일 | 졸업년월일 | 학교명 | 학위명 | 전공명 |
|---|---|---|---|---|---|
| | 평량평균 (G.P.A) | 국가 | 학위논문명 | | 지도교수명 |
| 학사 | 2013.03.04 | 2016.02.19 | 한양대학교 | 공학사 | 건축공학부 |
| | 4.23 / 4.5 | 대한민국 | | | |
| 석박사 | 2016.03.02 | 2022.02.18 | 한양대학교 | 공학박사 | 건축공학과 |
| | 4.39 / 4.5 | 대한민국 | 도시 사운드스케이프 경험의 시청각 상호작용을 통한 정신생리학적 회복 | | 전진용 |

## 3. 경력사항

| 기 간 | 근 무 년 월 | 근 무 기 관 명 | 직 위 | 타대학 전임교원여부 | 트랙구분 |
|---|---|---|---|---|---|
| 2022.08.16 ~ 2026.08.31 | 4 년 0 월 | 현대자동차 연구소 | 책임연구원 (정규직) | N | |
| 2022.03.01 ~ 2022.08.07 | 년 5 월 | 한국건설기술연구원(KICT) | 박사후연구원 (비정규직) | N | |
| | | | | | |

## 4. 정부기관 연구비수혜실적

| 수혜기관명 | 연구 책임자 | 참여 연구자수 | 참여형태 | 총 연구비 | 본인지분 연구비 | 수혜기간 |
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## 5. 산업체 연구비수혜실적

| 수혜기관명 | 연구 책임자 | 참여 연구자수 | 참여형태 | 총 연구비 | 본인지분 연구비 | 수혜기간 |
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## 6. 사회봉사실적

| 기 간 | 활 동 내 용 | 직위 또는 역할 |
|---|---|---|
| | | |

## 7. 특허보유실적

| 특허구분 | 특허명 | 국내/외 | 등록국가 | 등록일자 | 등록번호 |
|---|---|---|---|---|---|
| 특허 | 착용형 디스플레이 및 모바일 기기를 위한 삼차원 음향 캡처링 마이크 어레이 | 국내 | 대한민국 | 2018.03.28 | 10-1844822 |
| | | | | | |
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## 8. 기술이전

| 기술명 | 이전업체명 | 계약기간 | 기술이전액 | 본인지분율 | 본인지분액 |
|---|---|---|---|---|---|
| 공간 임펄스 응답을 활용한 데이터 증강을 통한 딥 | (주)한양에스엔에이 | 2022.01.20 ~ 2022.01.20 | 50,750,000 | 10 | 5,075,000 |
| | | | | | |
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## 9. 자격사항

| 취득일자 | 자격명 | 자격증번호 | 취득기관 |
|---|---|---|---|
| 2016.12.02 | LEED AP®Building Design + Construction | 11100929-AP-BD+C | U.S. Green Building Council |
| 2017.01.31 | WELL AP (미국 WELL 친환경 빌딩인증 전문가) | WELL-AP-000000853 | International WELL Building |
| 2017.02.21 | LEED AP®Interior Design + Construction | 11100929-AP-ID+C | U.S. Green Building Council |
| 2020.07.07 | 데이터분석 준전문가 (ADsP) | ADsP-025346 | 한국데이터산업진흥원장 (국가공 |

## 10. JA겸직가능학과

| 캠퍼스 | 대학 | 학과 | 전공 |
|---|---|---|---|
| 서울 | 공과대학 | 건축학부 | |
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## 11. 국제활동(연구) 실적

| (공동연구) 기관명 | (공동연구) 교수(연구자)명 | 기간 | 내용 |
|---|---|---|---|
| Sorbonne University | Jean-Dominique Polack | 2018.07.-2020.12. | 한-프랑스 국가 간 도시 사운드스케이프 인식 비교<br>“Urban soundscapes in the imaginaries of native digital users: guidelines for soundscape design”, Sustainability. Vol. 14(2), pp. 632. (2022)<br>Doi: https://doi.org/10.3390/su14020632 |
| University College London | Jian Kang | 2020.06.-2022.05. | 사운드스케이프 국제 표준 제정을 위한 국가간 사운드스케이프 평가 척도 비교<br>“Soundscape descriptors in eighteen languages: Translation and validation through listening experiments”, Applied Acoustics, Vol. 224, pp. 110109. (2024)<br>Doi: https://doi.org/10.1016/j.apacoust.2024.110109 |
| | | | |

## 12. 국제활동(연구) 계획

| (공동연구) 기관명 | (공동연구) 교수(연구자)명 | 시작예정기간 | 내용 | 진행률 |
|---|---|---|---|---|
| University College | Jian Kang | 2026.08.01 | 실내 사운드스케이프 디자인 연구 (미래 모 | 공동 연구 설계 |

| (공동연구) 기관명 | (공동연구) 교수(연구자)명 | 시작예정기간 | 내용 | 진행률 |
|---|---|---|---|---|
| | | | 빌리티 적용) | 단계 |
| | | | | |
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## 13. 국제활동(교육) 실적

| (공동교육) 기관명 | (공동교육) 교수(연구자)명 | 기간 | 내용 |
|---|---|---|---|
| RWTH Aachen University | Michael Vorlaender | 2016 | VR 입체음향 (3D 입체음향 시스템) 관련 정기 세미나를 통한 스피커/헤드폰 기반 입체음향시스템 구현 방법론 습득 (SCI 논문 게재 완료 및 박사 학위 심사위원 참여) |
| University College London | Jian Kang | 2018-2021 | 주기적인 컨퍼런스 미팅 및 포럼을 통한 사운드스케이프 연구 노하우 전수 및 공동 연구 진행 (박사 학위 심사위원 참여) |
| | | | |

## 14. 국제활동(교육) 계획

| (공동교육) 기관명 | (공동교육) 교수(연구자)명 | 시작예정기간 | 내용 | 진행률 |
|---|---|---|---|---|
| University College London | Jian Kang | 20270901 | 대학원생 공동 연구 지도 | 공동 연구 설계 단계 |
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## 15. 산학(기관)협력 실적

| 협력 (예정) 기관명 | 협력기간 | 내용 |
|---|---|---|
| 서울대학교 기계공학과 (현대자동차 산학협동) | 1년 (2026-2027) | 서울대학교-현대자동차 산학협동연구 주관 (과제 책임자)<br>└ 소비자 감성 중심 로드노이즈 고주파 개선을 위한 차체/흡차음 개선 연구<br>└ 차체 구조전달음 및 소음 공기전달음 제어 방안에 대한 시험/해석 검증 |
| 충남대학교 건축공학과 (현대자동차 산학협동) | 3년 (2024-2027) | 충남대학교-현대자동차 산학협동연구 주관 (과제 책임자)<br>└ 전기차 AVAS 경고음에 대한 운전자/보행자 인지 측면 사운드스케이프 디자인 연구<br>└ 소비자 만족도 증진을 위한 인간 감성 기반 AVAS 사운드스케이프 평가 지표 및 설계 가이드라인 제시 |
| 퍼시스 | 6개월 (2020) | 제품 및 공간조건에 따른 음환경 분석을 통한 오피스 제품 및 공간 플래닝 가이드 제안<br>└ 실내 공간의 시청각 요인 변화에 따른 사운드스케이프(음환경) 인식 변화 조사 및 적정 설계 가이드라인 제시 |
| 현대자동차 (선행연구팀) | 1년 (2021) | 차량 실내 음환경 개발을 위한 사용자 환경별 Perception 모델 개발<br>└ 차량 음환경에 대한 심리적 • 생리적 반응을 기반으로 한 실내 사운드스케이프의 사용자 반응 평가<br>└ 차량 내/외부 (풍경, 주광, 환경소음 등)컨텍스트를 반영한 종합적인 사운드스케이프 인식 모델 개발<br>└ 사용자 개인 맞춤형 음환경 서비스 제공을 위한 인식모델 개선 |

## 16. 산학(기관)협력 계획

| 협력 (예정) 기관명 | 시작예정기간 | 내용 | 진행률 |
|---|---|---|---|
| 현대자동차 | 20260801 | 현대자동차 근무 경험 및 산학 연구 주관 경험을 기반으로 장기 산학 협력 연구 수주 계획<br>└ 차량 음환경에 대한 정신생리학적 인지 반응 연구 (주행 인지, 경고음 디자인, 가상엔진사운드 디자인, 멀미 개선 등) | 산학(기관)협력 설계 단계 |
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| 17. 수상실적 | | | |
|---|---|---|---|
| 수상명 | 수여기관 | 수상일자 | 비고 |
| 현대자동차그룹 학술대회 특별상 | 현대자동차 | 2025.10.13 | |
| 유망과학자 감사장 | 한국소음진동공학회 | 2022.11.03 | |
| EAA Best Paper and Presentation Award | 유럽음향학회(EAA) | 2019.09.13 | 음향 분야 세계 최대 학회 (3년 주기 개최) |
| 20 Young Professional Congress Attendance Grants (YP Grants) | International Institute of Noise Control Engineering (I-INCE) | 2020.08.23 | |
| 박사학위 논문 우수상 | 한양대학교 | 2021.12.02 | |
| 연구실적 우수자상 (논문우수부문) | 한양대학교 | 2021.03.05 | |
| 연구실적 우수자상 (논문우수부문) | 한양대학교 | 2020.03.10 | |
| LH 국민 아이디어 공모전 (장려상 - LH 사장상) | LH 주택공사 | 2020.11.02 | |
| NSV 연구장려금 | 한국소음진동공학회 | 2022.06.04 | |

위 기재사항은 사실과 다름이 없습니다.

2026 년 03 월 12 일

지 원 자 조현인 (인)

**한양대학교 총 장** 귀하

# 연 구 실 적 목 록

| | | | |
|---|---|---|---|
| 소 속 | 서울 공과대학 | | |
| 지원분야 | 지능형 건축환경 시스템 | 연구분야 | 공학 > 건축공학 > 건축환경/설비 및 에너지 > 건축환경 |
| 성 명 | 조현인 | | |

| 번호 | 논문제목 | 발표일자 | 참여자명 | | | 학술지명 |
|---|---|---|---|---|---|---|
| | Category | 참여자수 / 역할 | 논문구분 / Index | I.F.<br>E.F. | I.F. 보정 | 카테고리별<br>상위% |
| 1 | Psycho-physiological Restoration with Audio-visual Interaction in Urban Soundscape Experience | 2022.02.18 | | | | |
| | | / | /<br>(학위논문) | | | % |
| 2 | Psycho-physiological restoration with audio-visual interactions through virtual reality simulations of soundscape and landscape experiences in urban, waterfront, and green environments | 2023.12.01 | Jeon, Jin Yong; Jo, Hyun In; Lee, Kounseok | | | SUSTAINABLE CITIES AND SOCIETY |
| | CONSTRUCTION & BUILDING TECHNOLOGY,ENERGY & FUELS,GREEN & SUSTAINABLE SCIENCE & TECHNOLOGY | 3/교신저자 | 국제적저명학술지 / SCI-E<br>(대표논문) | 10.696 | 1.275 | 2.9% |
| 3 | Compatibility of quantitative and qualitative data-collection protocols for urban soundscape evaluation | 2021.11.01 | Jo, Hyun In; Jeon, Jin Yong | | | SUSTAINABLE CITIES AND SOCIETY |
| | CONSTRUCTION & BUILDING TECHNOLOGY,ENERGY & FUELS,GREEN & SUSTAINABLE SCIENCE & TECHNOLOGY | 2/교신저자 | 국제적저명학술지 / SCI-E<br>(대표논문) | 5.268<br>0.0101 | 0.961 | 7.9% |
| 4 | Overall environmental assessment in urban parks: Modelling audio-visual interaction with a structural equation model based on soundscape and landscape indices | 2021.10.15 | Jo, Hyun In; Jeon, Jin Yong | | | BUILDING AND ENVIRONMENT |
| | CONSTRUCTION & BUILDING TECHNOLOGY,ENGINEERING, CIVIL,ENGINEERING, ENVIRONMENTAL | 2/제1저자 | 국제적저명학술지 / SCI-E<br>(대표논문) | 4.971<br>0.02522 | 1.163 | 3.0% |
| 5 | Potential restorative effects of urban soundscapes: Personality traits, temperament, and perceptions of VR urban environments | 2021.10.01 | Jeon, Jin Yong; Jo, Hyun In; Lee, Kounseok | | | LANDSCAPE AND URBAN PLANNING |
| | ECOLOGY,GEOGRAPHY, PHYSICAL | 3/교신저자 | 국제적저명학술지 / SCI-E<br>(대표논문) | 5.441<br>0.01601 | 1.095 | 8.0% |

# 연 구 실 적 목 록

| 소 속 | 서울 공과대학 | | |
|---|---|---|---|
| 지원분야 | 지능형 건축환경 시스템 | 연구분야 | 공학 > 건축공학 > 건축환경/설비 및 에너지 > 건축환경 |
| 성 명 | 조현인 | | |

| 번호 | 논문제목 | 발표일자 | 참여자명 | | | 학술지명 |
|---|---|---|---|---|---|---|
| | Category | 참여자수 / 역할 | 논문구분 / Index | I.F. E.F. | I.F. 보정 | 카테고리별 상위% |
| 6 | Quantification of Visual Attention by Using Eye-Tracking Technology for Soundscape Assessment Through Physiological Response | 2024.11.07 | Jo, Hyun In; Jeon, Jin Yong | | | International Journal of Environmental Research and Public Health |
| | | 2/제1저자 | 기타국제학술지 / SCOPUS | | | % |
| 7 | Soundscape descriptors in eighteen languages: Translation and validation through listening experiments | 2024.09.05 | Aletta, Francesco et al. | | | APPLIED ACOUSTICS |
| | ACOUSTICS | 57/참여자 | 국제적저명학술지 / SCI-E | 3.4 | 0.611 | 22.6% |
| 8 | Optimizing sound diffusion in a concert hall using scale-model measurements and simulations | 2022.06.01 | Jo, Hyun In; Jeon, Jin Yong | | | JOURNAL OF BUILDING ENGINEERING |
| | CONSTRUCTION & BUILDING TECHNOLOGY,ENGINEERING, CIVIL | 2/제1저자 | 국제적저명학술지 / SCI-E | 5.318 | 0.961 | 9.4% |
| 9 | Effect of noise sensitivity on psychophysiological response through monoscopic 360 video and stereoscopic sound environment experience: a randomized control trial | 2022.03.16 | Jo, Hyun In; Lee, Kounseok; Jeon, Jin Yong | | | SCIENTIFIC REPORTS |
| | MULTIDISCIPLINARY SCIENCES | 3/제1저자 | 국제적저명학술지 / SCI-E | 4.38 | 0.465 | 21.3% |
| 10 | Classification of sound environment based on subjective response with speech privacy in open plan offices | 2022.02.28 | Jo, Hyun In; Santika, Beta Bayu; Lee, Haram; Jeon, Jin Yong | | | APPLIED ACOUSTICS |
| | ACOUSTICS | 4/제1저자 | 국제적저명학술지 / SCI-E | 2.639 | 0.495 | 35.5% |
| 11 | Perception of urban soundscape and landscape using different visual environment reproduction methods in virtual reality | 2022.01.15 | Jo, Hyun In; Jeon, Jin Yong | | | APPLIED ACOUSTICS |
| | ACOUSTICS | 2/제1저자 | 국제적저명학술지 / SCI-E | 2.639 | 0.495 | 35.5% |

# 연 구 실 적 목 록

| 소 속 | 서울 공과대학 | | |
|---|---|---|---|
| 지원분야 | 지능형 건축환경 시스템 | 연구분야 | 공학 > 건축공학 > 건축환경/설비 및 에너지 > 건축환경 |
| 성 명 | 조현인 | | |

| 번호 | 논문제목 | 발표일자 | 참여자명 | | | 학술지명 |
|---|---|---|---|---|---|---|
| | Category | 참여자수 / 역할 | 논문구분 / Index | I.F. E.F. | I.F. 보정 | 카테고리별 상위% |
| 12 | Influence of indoor soundscape perception based on audiovisual contents on work-related quality with preference and perceived productivity in open-plan offices | 2022.01.15 | Jo, Hyun In; Jeon, Jin Yong | | | BUILDING AND ENVIRONMENT |
| | CONSTRUCTION & BUILDING TECHNOLOGY,ENGINEERING, CIVIL,ENGINEERING, ENVIRONMENTAL | 2/제1저자 | 국제적저명학술지 / SCI-E | 6.456 | 1.166 | 4.3% |
| 13 | Urban Soundscapes in the Imaginaries of Native Digital Users: Guidelines for Soundscape Design | 2022.01.01 | Polack, Jean-Dominique; Taupin, Philippe; Jo, Hyun In; Jeon, Jin Yong | | | SUSTAINABILITY |
| | ENVIRONMENTAL SCIENCES,GREEN & SUSTAINABLE SCIENCE & TECHNOLOGY | 4/공저자(이공계열은 수학분야 한정) | 국제적저명학술지 / SCI-E | 3.251 | 0.359 | 44.4% |
| 14 | Crossed effects of audio-visual environment on indoor soundscape perception for pleasant open-plan office environments | 2022.01.01 | Jeon, Jin Yong; Jo, Hyun In; Santika, Beta Bayu; Lee, Haram | | | BUILDING AND ENVIRONMENT |
| | CONSTRUCTION & BUILDING TECHNOLOGY,ENGINEERING, CIVIL,ENGINEERING, ENVIRONMENTAL | 4/교신저자 | 국제적저명학술지 / SCI-E | 6.456 | 1.166 | 4.3% |
| 15 | Urban soundscape categorization based on individual recognition, perception, and assessment of sound environments | 2021.12.01 | Jo, Hyun In; Jeon, Jin Yong | | | LANDSCAPE AND URBAN PLANNING |
| | ECOLOGY,GEOGRAPHY, PHYSICAL | 2/제1저자 | 국제적저명학술지 / SCI-E | 5.441 0.01601 | 1.095 | 8.0% |
| 16 | Diagnosis of Pneumonia by Cough Sounds Analyzed with Statistical Features and AI | 2021.11.01 | Chung, Youngbeen; Jin, Jie; Jo, Hyun In; Lee, Hyun; Kim, Sang-Heon; Chung, Sung Jun; Yoon, Ho Joo; Park, Junhong; Jeon, Jin Yong | | | SENSORS |
| | CHEMISTRY, ANALYTICAL,ENGINEERING, ELECTRICAL & ELECTRONIC,INSTRUMENTS & INSTRUMENTATION | 9/공저자(이공계열은 수학분야 한정) | 국제적저명학술지 / SCI-E | 3.275 0.08401 | 0.646 | 23.4% |

# 교육(강의)계획서

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| 지원소속 | 서울 공과대학 건축공학부 |
| 초빙분야 | 지능형 건축환경 시스템 |
| 성 명 | 조현인 |
| 내 용 | 한양대학교 건축공학과의 교육 목표는 실용학풍을 바탕으로 사회와 산업에 기여할 수 있는 창의적 건축 인재를 양성하는 것입니다. 저는 건축환경 연구와 산업 경험을 바탕으로 이론, 데이터, 설계가 결합된 융합형 교육을 실현하고자 합니다. 특히 건축음향, 환경 설계, 스마트 건축환경 시스템 등 융합적 교과목을 중심으로 학생들이 실제 문제를 해결하는 프로젝트 기반 학습(Project-Based Learning) 중심의 교육을 추진할 계획입니다.<br><br>우선 기존 교과목인「건축환경공학」,「건축음향」,「환경과생태건축」등을 기반으로 데이터 기반 건축환경 교육을 강화하고자 합니다. 최근 건축환경 분야는 센서 데이터와 AI 분석 기술을 활용한 환경 관리 연구가 확대되고 있으며 이러한 변화에 대응하기 위해 학생들이 데이터 기반 환경 설계 능력을 갖추는 것이 중요합니다. 이를 위해 IoT 센서를 활용하여 실내외 환경 데이터를 측정하고 데이터 분석 및 시뮬레이션 도구를 활용하여 환경 문제를 분석하는 실습형 수업을 운영할 계획입니다. 학생들은 환경 데이터를 수집하고 분석하여 건축 설계에 반영하는 과정을 경험함으로써 데이터 기반 환경 설계 사고를 습득할 수 있습니다.<br><br>또한 산업 연계 교육과 캡스톤 디자인을 적극적으로 운영하고자 합니다. 현대자동차 연구소에서의 NVH 연구 경험과 한국건설기술연구원에서의 스마트 건축 연구 경험을 활용하여 산업과 연계된 실무 중심 프로젝트 수업을 운영할 계획입니다. 예를 들어 친환경 주거단지 음환경 개선, 스마트시티 공공공간 환경 설계, 전기차 충전 인프라의 환경 설계 등 실제 도시 및 산업 문제를 해결하는 캡스톤 프로젝트를 수행하도록 지도할 예정입니다. 이러한 프로젝트 기반 수업을 통해 학생들은 기술적 분석 능력과 창의적 설계 능력을 동시에 기를 수 있을 것입니다.<br><br>또한 건축환경 분야의 특성을 고려하여 실험과 데이터 분석 중심의 교육 환경을 구축하고자 합니다. 학생들이 직접 환경 데이터를 측정하고 분석할 수 있도록 간단한 환경 센서 및 분석 도구를 활용한 실습 프로그램을 운영하고, 이를 연구 프로젝트와 연계하여 학부생들도 실제 연구 과정에 참여할 수 있도록 지도할 계획입니다. 이를 통해 학생들이 연구 경험을 조기에 접하고 건축환경 분야에 대한 이해를 높일 수 있도록 지원하고자 합니다.<br><br>글로벌 및 융합 교육 역시 강화하고자 합니다. 해외 대학과의 공동 연구 경험을 바탕으로 국제 공동 워크숍 및 세미나를 추진하고 수업을 영어로 운영하여 학생들의 글로벌 역량을 높일 계획입니다. 또한 학생들이 국제 학회나 디자인 공모전에 참여할 수 있도록 지도하고 이를 포트폴리오와 연계하여 연구 경험과 실무 경험을 동시에 쌓을 수 있도록 지원하겠습니다.<br><br>마지막으로 멘토링 중심의 학생 지도 체계를 구축하고자 합니다. 학생들의 진로 방향은 연구, 설계, 산업, 공공 정책 등 다양하게 나타나므로 학생 개개인의 관심과 역량에 맞는 맞춤형 지도를 제공할 계획입니다. 건축환경 분야는 기술과 인간 경험이 결합되는 영역이기 때문에 단순한 지식 전달을 넘어 문제 해결 능력과 윤리 의식을 함께 함양하는 교육이 중요합니다. 한양대학교의 교육 철학인 '사랑의 실천'을 바탕으로 학생들이 사회와 환경을 이해하고 지속가능한 미래를 설계할 수 있는 창의적 건축 인재로 성장할 수 있도록 지도하겠습니다. |

| 강의가능과목 | |
|---|---|
| 1 | 건축환경공학 |
| 2 | 건축음향 |
| 3 | 환경과생태건축 |
| 4 | 확률통계론 |
| 5 | 스마트 건축환경 시스템 |
| 6 | 건축 빅데이터 분석 |
| 7 | 건축 환경 디자인 |

# 연구계획서

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| 지원소속 | 서울 공과대학 건축공학부 |
| 초빙분야 | 지능형 건축환경 시스템 |
| 성 명 | 조현인 |
| 내 용 | 한양대학교 건축공학과는 실용학풍을 기반으로 첨단 기술과 산업을 연결하는 연구 중심 학과로, 스마트시티와 지속가능 건축환경 분야에서 활발한 연구가 이루어지고 있습니다. 저는 이러한 연구 철학에 공감하며 건축음향 및 환경설계를 기반으로 한 지능형 건축환경 시스템(Intelligent Built Environment System) 연구를 수행하고자 합니다. 특히 도시와 건축, 그리고 모빌리티 환경을 통합적으로 고려한 데이터 기반 멀티스케일 음환경 분석 및 설계 기술을 핵심 연구 주제로 설정하고, 센서 데이터와 인공지능 기술을 활용한 건축환경 연구를 추진할 계획입니다.<br><br>본 연구의 목표는 AI 기반 건축환경 데이터 분석과 사운드스케이프 설계 모델 구축입니다. 건축환경은 건물 내부, 건물군, 도시 공간 등 다양한 규모에서 형성되며, 최근에는 IoT 센서와 데이터 기반 기술을 통해 환경 상태를 실시간으로 분석하고 관리하는 지능형 환경 시스템으로 발전하고 있습니다. 이에 따라 본 연구에서는 건축물 내부(마이크로 스케일), 건물군 및 캠퍼스 환경(메조 스케일), 도시 공간(매크로 스케일)에 이르는 다양한 공간 규모에서 음환경 데이터를 수집하고 인간의 심리적 쾌적성과 연계된 통합 음환경 평가 모델을 개발하고자 합니다. 이를 통해 기존의 소음 규제 중심 접근을 넘어 데이터 기반 환경 설계와 지능형 환경 관리 기술을 구현하는 것을 목표로 합니다.<br><br>첫 번째 연구 단계에서는 AI 기반 건축환경 데이터 분석 및 음환경 진단 기술을 개발하고자 합니다. IoT 기반 음향 센서를 활용하여 실내·외 환경 데이터를 수집하고 머신러닝 기반 분석 기법을 통해 공간 유형별 소음 특성과 음환경 패턴을 분석할 계획입니다. 특히 한국건설기술연구원에서 수행했던 AI 기반 건축에너지 진단 연구 경험을 활용하여 건축환경 데이터 분석 기술을 음환경 연구로 확장하고자 합니다. 이를 통해 강의실, 도서관, 공공시설 등 다양한 공간 유형에서 음환경을 진단하고 공간 유형별 최적의 음환경 설계 가이드라인을 제시할 계획입니다.<br><br>두 번째 연구 단계에서는 모빌리티와 도시 환경의 상호작용을 고려한 음환경 연구를 수행하겠습니다. 최근 전기차 및 자율주행차의 확산으로 도시 음환경이 크게 변화하고 있으며 이에 따른 보행자 안전과 도시 환경 품질 문제가 새롭게 제기되고 있습니다. 현대자동차 연구소에서 수행했던 전기차 AVAS 및 NVH 연구 경험을 바탕으로 전기차 및 미래 모빌리티가 도시 음환경에 미치는 영향을 분석하고 이를 도시 설계 및 도로 인프라 계획에 반영할 수 있는 기술적 근거를 제시하고자 합니다.<br><br>마지막으로 도시 및 건축 환경에서 수집되는 데이터를 활용하여 디지털 트윈 기반 스마트 음환경 관리 플랫폼 구축 연구를 수행하겠습니다. 이를 통해 도시 전반의 음환경 상태를 모니터링하고 AI 기반 환경 예측 모델을 통해 공간 환경을 능동적으로 관리할 수 있는 기술을 개발하고자 합니다. 이러한 연구는 스마트시티 환경 관리 플랫폼과 연계되어 건축환경 연구를 데이터 기반 도시 환경 관리 기술로 확장하는 기반이 될 것입니다.<br><br>본 연구는 건축환경 연구에 AI 기반 데이터 분석과 인간 중심 환경 설계 개념을 결합한 연구 방향을 제시하는 것을 목표로 합니다. 또한 산업체 및 해외 대학과의 협력 연구를 통해 연구 성과를 실제 도시 환경 문제 해결로 확장하고자 합니다. |

※ 작성요령

1) 향후 3~5년간 계획을 1페이지 이내로 한글 또는 영문으로 작성

2) 교육계획(강의, 교과개발, 교재개발 등) 작성 시, 범위와 내용의 제한은 없으나 구체적으로 기술

# 논 문 (작품/전시, 연주) 개 요

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| 소 속 | 서울 공과대학 |
| 지원분야 | 지능형 건축환경 시스템 |
| 성 명 | 조현인 |

| 번호 | 논문구분<br>Index | 논 문 제 목 | 학술지명<br>발표일 |
|---|---|---|---|
| 1 | 학위논문 | Psycho-physiological Restoration with Audio-visual Interaction in Urban Soundscape Experience | 2022.02.18 |

Urban noise has increasingly affected the quality of life and health of city residents, raising the need for healthier urban environments. With the emergence of the soundscape concept, which views urban sounds as environmental resources, research on urban sound perception has expanded. However, studies examining the restorative effects of soundscape experiences remain limited. This study investigated the psycho-physiological restoration effects of soundscape experiences in various urban environments, focusing on audio-visual interactions, and proposed a soundscape restoration model that can guide sustainable urban design.

Chapter 1 examined the compatibility of soundscape evaluation methods suggested in ISO 12913-2 and compared their advantages to identify appropriate tools for evaluating restoration effects. It also verified the ecological validity of virtual reality (VR) experimental environments by comparing soundwalk responses in real environments with laboratory-based VR evaluations.

Chapter 2 analyzed the influence of audio-visual interaction on soundscape perception in different urban environments. The relative contributions of visual and auditory information to satisfaction were examined. In addition, quantitative indicators based on perceived proportions of natural, traffic, and human sounds were proposed to support urban park design, and design guidelines were derived through structural equation modeling.

Chapter 3 explored the psycho-physiological restoration effects of soundscape experiences using psychological responses (questionnaires) and physiological indicators such as heart rate variability and EEG. Restoration effects were compared across urban and natural environments including forests and seas. Based on these results, a predictive restoration model for urban soundscapes was developed to support the creation of healthier and more sustainable urban environments.

# 논 문 (작품/전시, 연주) 개 요

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| 소 속 | 서울 공과대학 |
| 지원분야 | 지능형 건축환경 시스템 |
| 성 명 | 조현인 |

| 번호 | 논문구분<br>Index | 논 문 제 목 | 학술지명<br>발표일 |
|---|---|---|---|
| 2 | 국제적저명학술지<br>SCI-E | Psycho-physiological restoration with audio-visual interactions through virtual reality simulations of soundscape and landscape experiences in urban, waterfront, and green environments | SUSTAINABLE CITIES AND SOCIETY<br>2023.12.01 |

The literature lacks a model of the specific environmental element that induces restoration. Thus, this study contributes to the development of sustainable and healthy environments by examining the effect of the soundscape and landscape experience on psycho-physiological restoration from the audio-visual interaction perspective. It examined various urban and natural environments to propose a restoration model for environmental design application. Sixty participants evaluated three each of urban, waterfront, and green environments using virtual reality technology. The audio-visual characteristics of environments were quantified, and environmental perception was assessed per 14 audio-visual element identifications, six perceived affective qualities, and six overall environmental qualities. The restoration responses were classified as psychological and physiological. Accordingly, 10 restoration models that can interpret design factors of the urban environment with a maximum correlation of 0.80 and an explanatory power of 22 % emerged from urban audio-visual design elements and psycho-physiological responses. Consequently, to improve restoration response, visually attractive and spatialnatural landscapes and acoustically supportive and tranquil soundscapes should be created. Moreover, urban designers can use the models to clearly identify physical characteristics that should be prioritized when promoting the health improvement of urbanites. Furthermore, the study findings can serve as basic data for the development of healthy environments.

# 논 문 (작품/전시, 연주) 개 요

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| 소 속 | 서울 공과대학 |
| 지원분야 | 지능형 건축환경 시스템 |
| 성 명 | 조현인 |

| 번호 | 논문구분<br>Index | 논 문 제 목 | 학술지명<br>발표일 |
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| 3 | 국제적저명학술지<br>SCI-E | Compatibility of quantitative and qualitative data-collection protocols for urban soundscape evaluation | SUSTAINABLE CITIES AND SOCIETY<br>2021.11.01 |

This study investigates the compatibility of soundscape evaluation results based on the data-collection protocols proposed in ISO 12913-2, Method A (questionnaire), Method B (questionnaire and open-answer), and Method C (narrative interview), as well as guidelines for evaluating urban soundscapes. We assessed the soundscapes of 10 multi-functional sites in an urban environment through the responses of 50 participants to questions on sound source identification, perceived affective quality, and overall quality. Using virtual reality technology, we reproduced a laboratory environment similar to the actual assessment sites. The responses to sound source identification were similar for each protocol. Regarding perceived affective quality, the pleasantness-eventfulness model derived from Method A was also found in the text-mining results of Methods B and C; additional emotional responses were discovered. Regarding overall quality, the preference for each assessment site was similar for each protocol. Method C revealed the influence of non-acoustic factors on soundscape perception. The quantitative data protocol was appropriate for a large group and for deriving a generalized model, whereas the qualitative data protocol was effective for small groups or an in-depth analysis of certain sites. The findings of this study are useful for assessing urban soundscapes and informing urban planning.

# 논 문 (작품/전시, 연주) 개 요

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| 4 | 국제적저명학술지<br>SCI-E | Overall environmental assessment in urban parks: Modelling audio-visual interaction with a structural equation model based on soundscape and landscape indices | BUILDING AND ENVIRONMENT<br>2021.10.15 |

This study developed guidelines for the design of park environments for urban residents by conducting an overall environmental assessment from the perspective of audio-visual interaction. We collected information about audio-visual stimuli at 18 locations in three urban parks and created an assessment environment within a laboratory using virtual reality. A subjective assessment questionnaire was administered to 30 participants regarding audio-visual elements, perceived affective quality, and overall environmental assessment. Further, soundscape and landscape indices were defined using the audio-visual perception ratio between natural-related elements and traffic/human-related elements. Based on the subjective assessment responses, an environmental perception model using a structural equation model was developed. The results demonstrated that pleasantness should be prioritized over eventfulness. Water-related elements are efficient for increasing pleasantness, while providing greenery-related elements efficiently controls eventfulness. We found that audio-visual satisfaction improves when there is a higher share of water- and greenery-related elements than traffic-related elements. However, regarding the visual perception ratio of the human presence to natural elements, water features were found to be effective as a design factor for visual satisfaction and greenery for sound satisfaction. Considering the importance of human-related elements, we confirmed that designs are needed that will induce an influx of people into parks at an appropriate level to enhance their overall environmental satisfaction. This study's findings are expected to serve as design guidelines and supporting data to create pleasant urban park environments.

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| 5 | 국제적저명학술지<br>SCI-E | Potential restorative effects of urban soundscapes: Personality traits, temperament, and perceptions of VR urban environments | LANDSCAPE AND URBAN PLANNING<br>2021.10.01 |

This study investigated the potential restorative (PR) effects of urban soundscapes. To establish a new set of PR criteria for urban environments, this study recreated 10 urban sites in virtual reality, instead of using the typical method of questionnaires. A list of restoration-related semantic terms was derived from narrative interviews of 50 subjects who freely expressed their perceptions of the space. PR criteria were determined based on the derived list and two groups were found: the PR group and potential non-restorative group. The objective characteristics of urban soundscapes-acoustic and visual characteristics and characteristics of moving objects-were analyzed. Participants provided their subjective responses regarding sound source identification, perceived affective quality, and overall quality. The Temperament and Character Inventory-Revised-Short Version and Satisfaction with Life Scale were utilized to analyze participants' individual characteristics. Based on it, a model was developed to predict whether PR criteria could be satisfied, which was 82.6% accurate (85.9% specificity, 71.8% sensitivity). It was determined that the overall level of preference regarding an urban soundscape is a significant factor in satisfying PR criteria. In particular, the design and planning of urban spaces should aim to reduce a space's loudness and increase the revisitation rate to promote the restoration of the urban residents' mental health. Furthermore, for individuals, a trait based on acquired experience has a greater impact on the PR effects of urban soundscapes than innate temperament. This study's findings can serve as supporting data to design sustainable and health-promoting urban soundscapes.

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| 6 | 기타국제학술지<br>SCOPUS | Quantification of Visual Attention by Using Eye-Tracking Technology for Soundscape Assessment Through Physiological Response | International Journal of Environmental Research and Public Health<br>2024.11.07 |

Abstract: Because soundscapes affect human health and comfort, methodologies for evaluating them through physiological responses have attracted considerable attention. In this study, we proposed a novel method for evaluating visual attention by using eye-tracking technology to objectively assess soundscape perception. The study incorporated questionnaire surveys and physiological measurements focusing on visual attention responses. Results from the questionnaire indicated that perceptions of vehicles and the sky were 6% and 26% more sensitive, respectively, whereas perceptions of vegetation, based on physiological responses, were approximately 3% to 50% more sensitive. The soundscape quality prediction model indicates that the proposed methodology can complement conventional questionnaire-based models and provide a nuanced interpretation of eventfulness relationships. Additionally, the visual attention quantification technique enhanced the restoration responses of questionnaire-based methods by 1-2%. The results of this study are significant because the study introduces a novel methodology for quantifying visual attention, which can be used as a supplementary tool for physiological responses in soundscape research. The proposed method can uncover novel mechanisms of human perception of soundscapes that may not be captured by questionnaires, providing insights for future research in soundscape evaluation through physiological measurements.

# 논 문 (작품/전시, 연주) 개 요

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| 7 | 국제적저명학술지<br>SCI-E | Soundscape descriptors in eighteen languages: Translation and validation through listening experiments | APPLIED ACOUSTICS<br>2024.09.05 |

This paper presents the outcomes of the Soundscape Attributes Translation Project (SATP), an international initiative addressing the critical research gap in soundscape descriptors translations for cross-cultural studies. Focusing on eighteen languages - namely: Arabic, Chinese, Croatian, Dutch, English, French, German, Greek, Indonesian, Italian, Japanese, Korean, Malay, Portuguese, Spanish, Swedish, Turkish, and Vietnamese - the study employs a four-step procedure to evaluate the reliability and cross-cultural validity of translated soundscape descriptors. The study introduces a three-tier confidence level system (Low, Medium, High) based on adjusted angles, which are a measure proposed to correct the soundscape circumplex model (i.e., the pleasant-eventful space proposed in the ISO 12913 series) of a given language. Results reveal that most languages successfully maintain the quasi-circumplex structure of the original soundscape model, ensuring robust cross-cultural validity. English, Arabic, Chinese (Mandarin), Croatian, Dutch, German, Greek, Indonesian, Italian, Spanish, Swedish, and Turkish achieve a High confidence level. French, Japanese, Korean, Malay, Portuguese, and Vietnamese demonstrate varying confidence levels, highlighting the importance of the preliminary translation. This research significantly contributes to standardized cross-cultural methodologies in soundscape perception research, emphasizing the pivotal role of adjusted angles within the soundscape circumplex model in ensuring the accuracy of dimensions (i.e., attributes) locations. The SATP initiative offers insights into the complex interplay of language and meaning in the perception of environmental sounds, opening avenues for further crosscultural soundscape research.

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| 8 | 국제적저명학술지<br>SCI-E | Optimizing sound diffusion in a concert hall using scale-model measurements and simulations | JOURNAL OF BUILDING ENGINEERING<br>2022.06.01 |

To propose optimal guidelines for the diffusion design of concert halls, sound field variations in the presence or absence of reflectors (canopies and clouds) and diffusers were comprehensively examined in terms of stage and auditorium acoustics using 1:25 scale-model measurements and acoustic simulations. The sound field variations were investigated based on the just-noticeable differences. For stage acoustics, the stage support parameters STEarly and STLate were analyzed; for auditorium acoustics, the reverberation time (RT), early decay time (EDT), clarity (C-80), and center time (T-S) were analyzed. The canopies reduced the reverberations but increased STEarly and STLate.( )The clouds increaed STEarly and reduced the deviation of the auditorium EDT and T-S whilst increasing the initial EDT that had been reduced by the canopies. The diffusers decreased STEarly , STLate, RT, EDT, and T-S regardless of the attachment position, and they increased the C-80 of some auditoriums. However, diffuser attachments reduced the seat-to-seat deviations of acoustic parameters. Furthermore, attaching diffusers near to (as opposed to far from) the sound source represents the optimal design strategy for increasing the density of early reflections. This study is significant as an initial attempt to examine the effects of concert hall diffusion design in terms of both auditorium and stage acoustics, and it is significant in that it presents an appropriate method for evaluating the effects of a concert hall diffusion design that can be employed by architects and sound consultants. The findings of this study are expected to be suitable as basic data for the diffusion design of rectangular concert halls.

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| 9 | 국제적저명학술지<br>SCI-E | Effect of noise sensitivity on psychophysiological response through monoscopic 360 video and stereoscopic sound environment experience: a randomized control trial | SCIENTIFIC REPORTS<br>2022.03.16 |

Noise sensitivity is a crucial factor affecting subjective psychophysiological responses to the acoustic environment of various indoor and outdoor spaces. This study examines how noise sensitivity or hyperacusis affects the recovery of emotional and autonomic nervous system (ANS) responses when experiencing various monoscopic 360 video and stereoscopic sound environments (urban and natural) that represent the actual environment. A total of 60 general participants with mild depression, stress, and anxiety were examined using a survey to investigate individual characteristics, including noise sensitivity, and K-means clustering was used to classify them into sensitivity groups. Emotional and physiological responses were measured using the Korean edition of Profile of Mood States and by assessing heart rate variability, respectively. Overall, the emotional recovery effect was greater in the natural than the urban environment, and the homeostatic mechanism of the ANS was better maintained, thereby increasing stress resistance. Noise sensitivity did not have considerable effect on psychophysiological recovery in the natural environment, but had a significant effect on emotional response in the urban environment. This can be used as basic data in seeking customized emotional recovery for individuals using monoscopic 360 video and stereoscopic sound technology in the future.

# 논 문 (작품/전시, 연주) 개 요

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| 10 | 국제적저명학술지<br>SCI-E | Classification of sound environment based on subjective response with speech privacy in open plan offices | APPLIED ACOUSTICS<br>2022.02.28 |

To implement pleasant open plan offices (OPOs) in the preference aspect based on the subjective responses of the workers, we proposed novel acoustic environment design classification criteria for speech privacy-related parameters. To this end, we created acoustic stimuli through computer simulations and surveyed the related preference responses of test participants. Then, we selected classification criteria based on the survey response results. First, field measurements were performed in two typical OPOs, and a computer simulation model was built. Twelve different sound stimuli were generated by changing the material absorption coefficient conditions for the ceiling and floor inside an office. The investigation of the noise sensitivity and satisfaction of each sound source according to the individuals indicated that the A-weighted sound pressure level at 4 m (L-p,L-A,L-S,L-4m) and spatial decay rate of speech (D-2,D-S) were the major acoustical parameters that determined OPO satisfaction, with L-p,L-A,L-S,L-4m considered first in the OPO design stage. The L-p,L-A,L-S,L-4m class interval was set to 2 dB, and a new four-level classification criteria was proposed, in which the participant response rates of satisfaction were divided according to the change of L-p,L-A,L-S,L-4m into 25% intervals: A (< 46 dB), B (46 dB < x < 48 dB), C (48 dB < x < 50 dB), and D (> 50 dB). A large difference of two or more satisfaction classes was observed between the high and low noise sensitivity groups for the same L-p,L-A,L-S,L-4m, confirming that the noise sensitivity of workers influences the preferences for OPOs. The proposed classification criteria are expected to be applicable to the design of comfortable OPOs in the future.(c) 2021 Elsevier Ltd. All rights reserved.

# 논 문 (작품/전시, 연주) 개 요

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| 11 | 국제적저명학술지<br>SCI-E | Influence of indoor soundscape perception based on audiovisual contents on work-related quality with preference and perceived productivity in open-plan offices | BUILDING AND ENVIRONMENT<br>2022.01.15 |

Open-plan offices (OPOs) are being increasingly implemented to promote communication and collaboration among coworkers. We investigated the effects of indoor soundscape perception on productivity and preferences for the work environment in OPOs containing various audiovisual contents to achieve high work-related quality. A subjective evaluation was performed through two experiments conducted in a virtual reality environment with various stimuli. In Experiment 1, we investigated the correlations of visual complexity and acoustical variability with work-related quality. The results on a seven-point Likert-like scale (-3 to 3) indicated that visual complexity and maximum appropriate acoustical variability of an OPO was recommended at-1.3 to-0.2 and-0.3, respectively. In Experiment 2, the participants identified various noise sources and responded on their effects on the perceived affective quality and work-related quality. The results revealed that indoor soundscape perception can be interpreted in a comfort-content dimension and is significantly influenced by worker interactions such as conversation and laughter. Increased comfort improved the work environment preferences, and increased content significantly influenced the perceived productivity. Considering the factor score of the comfort-content dimensions, these results suggest the need to design a balanced OPO sound environment with preferred and productive zones by ensuring comfort and content distributions in the ranges of 0.3-0.7 and 0.6-1.2, respectively. The need to achieve an appropriate level of content in the indoor soundscape is confirmed, contradicting the notion that OPOs should be quiet. Therefore, these results can guide the design of sound en-vironments in open office spaces.

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| 12 | 국제적저명학술지<br>SCI-E | Perception of urban soundscape and landscape using different visual environment reproduction methods in virtual reality | APPLIED ACOUSTICS<br>2022.01.15 |

This study investigated the effects of visual environment reproduction methods of virtual reality (VR) on soundscape and landscape assessments for reproducing VR environments in laboratory settings. It also subjectively assessed urban environments to compare assessment method qualities. Specific monitor and head-mounted display (HMD) environments were selected as visual reproduction methods. Spatial audio that applied first-order ambisonic-based head-tracking technology was equally provided in both visual environments. We analyzed the subjective responses of 40 individuals according to overall satisfaction, VR reproduction quality, perceived audio-visual elements, and adjective attributes describing soundscape and landscape. The results showed that urban environments were more sensitively assessed through HMD than the monitors. Negative and positive components were more clearly perceived in the HMD environment. Furthermore, the negative effects of artificial components were rated highly in the HMD environment. The surrounded landscape affected urban contextual perceptions positively in HMD and negatively in monitor environments, making it an important assessment factor. Consequently, monitor and HMD environments were beneficial to the overall awareness of urban environments and spatial presence, respectively. The discoveries and data of this study could be extensively used as a basis for developing experimental methodologies to assess the soundscape and landscape of urban environments in laboratory settings. (C) 2021 Elsevier Ltd. All rights reserved.

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| 13 | 국제적저명학술지<br>SCI-E | Urban Soundscapes in the Imaginaries of Native Digital Users: Guidelines for Soundscape Design | SUSTAINABILITY<br>2022.01.01 |

We present an experiment run along the lines of a pilot experiment in China based on collages and narratives to illustrate the participants' experience with urban sceneries. Its aim was to develop a conceptual model based on narrative analysis that linked objects of the environment to the perceived properties of the soundscape. Participants in groups of two were free to clip any image they wanted from a selection of magazines-the same for all groups within each country-and free to add comments or drawings on their collages. Then, they had to present their collages to the other participants, and the presentations were recorded and transcribed. The structural semantic model that underlies the descriptions of the collages and narratives is presented. The results of the analysis were comparable with previous studies, since ideal urban environments should be calm, quiet and green, but urban environments should also promote cultural activities and the possibility to escape outside the city. The analysis also allowed for attaining the emotions created by soundscapes. Thus, the semantic model can be used as a conceptual model for a soundscape, from which guidelines for soundscape planning and design can be derived, as well as suggestions for innovative soundscapes.

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| 14 | 국제적저명학술지<br>SCI-E | Crossed effects of audio-visual environment on indoor soundscape perception for pleasant open-plan office environments | BUILDING AND ENVIRONMENT<br>2022.01.01 |

An open-plan office (OPO) layout promotes communication and teamwork among workers and provides flexibility in working; however, it also adversely affects work satisfaction, health, and worker privacy, leading to dissatisfaction and poor task performance. In this study, the subjective responses in an audio-visual OPO environment were analyzed to investigate the effects of the environment on indoor soundscape perception for designing pleasant OPOs. The outdoor environments of four OPOs with four different visual layouts were divided based on two conditions with different window views (good and bad views) to model eight visual stimuli. The auditory aspect was evaluated by computer simulation using three sound sources. For subjective assessment, virtual reality technology was employed with 24 audio-visual stimuli, and thirty-four subjects were asked to rate the perceived affective quality and overall quality. Task performance was quantified using auditory digit span tasks. The spatial preference decreased with the Loud, Variable, and Reverberant attributes and increased with the Bright, Orderly, and Wide attributes. For the same OPO soundscape, the window area ratio and floor area showed positive correlations with visual privacy, visual satisfaction, speech privacy, and sound satisfaction. However, work satisfaction and task performance exhibited contradictory relationships with the visual parameters. Furthermore, wide, open workspaces yielded high work satisfaction, whereas narrow, confined work spaces yielded high task performance. These results indicate that the relationship between work satisfaction and task performance must be considered for a balanced OPO design, providing practical guidance for designing pleasant OPO environments.

# 논 문 (작품/전시, 연주) 개 요

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| 15 | 국제적저명학술지<br>SCI-E | Urban soundscape categorization based on individual recognition, perception, and assessment of sound environments | LANDSCAPE AND URBAN PLANNING<br>2021.12.01 |

This study proposes soundscape recognition models by clustering people based on differences in sound source perceptions. We investigated the effect of sound source identification differences on urban soundscape perception by categorizing people's environmental sound recognition in outdoor environments. Virtual reality technology employing audio-visual stimuli collected in various urban environments replicated actual environments. Fifty participants' subjective responses regarding sound source identification, perceived affective quality (8 typical (ISO scale) and 116 extensive attributes (Swedish rating scale)), and overall quality were surveyed. Their categorizations by sound source identification were divided into three clusters: Cluster 1-Attentive to traffic noise and other noises, Cluster 2-Less attentive to the sound environment, and Cluster 3-Attentive to natural and human sounds. Even in identical spaces, participants identified different sound sources, as each cluster focused on different sounds. The soundscape perceptual components were derived differently for each cluster; Cluster 2 extracted additional perception dimensions, i.e., tranquil and relaxed soundscapes. The results showed that each sound source that received an attentive reaction had a positive effect on soundscape perception, showing that appropriate human activities can be encouraged to improve relaxation via soundscape enhancements. The overall quality assessment by cluster revealed similar results, but the resulting indicators' effects varied. The study's different soundscape recognition models for each cluster, based on the relationship between soundscape indicators and descriptors, present a new perspective for interpreting urban soundscape perception and can also be used effectively in urban planning design.

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| 16 | 국제적저명학술지<br>SCI-E | Diagnosis of Pneumonia by Cough Sounds Analyzed with Statistical Features and AI | SENSORS<br>2021.11.01 |

Pneumonia is a serious disease often accompanied by complications, sometimes leading to death. Unfortunately, diagnosis of pneumonia is frequently delayed until physical and radiologic examinations are performed. Diagnosing pneumonia with cough sounds would be advantageous as a non-invasive test that could be performed outside a hospital. We aimed to develop an artificial intelligence (AI)-based pneumonia diagnostic algorithm. We collected cough sounds from thirty adult patients with pneumonia or the other causative diseases of cough. To quantify the cough sounds, loudness and energy ratio were used to represent the level and its spectral variations. These two features were used for constructing the diagnostic algorithm. To estimate the performance of developed algorithm, we assessed the diagnostic accuracy by comparing with the diagnosis by pulmonologists based on cough sound alone. The algorithm showed 90.0% sensitivity, 78.6% specificity and 84.9% overall accuracy for the 70 cases of cough sound in pneumonia group and 56 cases in non-pneumonia group. For same cases, pulmonologists correctly diagnosed the cough sounds with 56.4% accuracy. These findings showed that the proposed AI algorithm has value as an effective assistant technology to diagnose adult pneumonia patients with significant reliability.